

Assignment 1

Exploratory Data Analysis & Feature Engineering

SCANIA Component X Dataset

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Course: AI & Neural Networks

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GitHub: <https://github.com/JaboDabo/SCANIA-Compenent-X-AI-NN->

Dataset: <https://researchdata.se/en/catalogue/dataset/2024-34>

1. Introduction

This report presents the results of an exploratory data analysis (EDA) and feature engineering performed on the SCANIA Component X dataset. The dataset is a real-world multivariate time series collected from anonymized engine components across a fleet of SCANIA heavy-duty trucks. It was introduced at the IDA 2024 Industrial Challenge at Stockholm University and published in Scientific Data (Nature, 2025). The dataset is publicly available under CC BY 4.0 license.

The objective is to understand the data structure, identify key patterns, handle missing values and outliers, and engineer new features for neural network models in subsequent assignments.

2. Dataset Description

The dataset comprises three types of files for each split (train 70%, validation 15%, test 15%):

2.1 Operational Readouts

Contains sporadic, unevenly sampled time series readouts from each vehicle. Each row is a unique readout at a specific time_step representing operational time units since Component X was installed. The full training file has 1,122,452 observations from 23,550 vehicles with 107 columns.

The 14 sensor variables are organized into two types:

- 6 histogram variables: 167 (10 bins), 272 (10 bins), 291 (11 bins), 158 (10 bins), 459 (20 bins), 397 (36 bins). Each histogram represents distributions across operating conditions (e.g., distance driven at different temperature ranges). Total: 97 columns.
- 8 numerical counters: 171_0, 666_0, 427_0, 837_0, 309_0, 835_0, 370_0, 100_0. Accumulative values that always increase over time. Total: 8 columns.

2.2 Time-to-Event (TTE)

Repair records with:

- length_of_study_time_step: total observation period
- in_study_repair: binary (1 = failed, 0 = no repair/censored)

Training set: 21,278 healthy (90.4%) vs 2,272 failed (9.6%).

2.3 Specifications

8 anonymized categorical features (Spec_0 to Spec_7) for truck configuration (engine type, wheel setup, etc.). Values: Cat0 to Cat28.

2.4 Label System

Training labels: BINARY (in_study_repair: 0/1)

Validation/test labels: 5-CLASS (0-4) based on time to failure:

- Class 0: >48 time units before failure (safe)
- Class 1: 24-48 time units before failure
- Class 2: 12-24 time units before failure
- Class 3: 6-12 time units before failure
- Class 4: 0-6 time units before failure (imminent)

The evaluation uses an asymmetric cost matrix where missing a failure (cost 200-500) is far more expensive than a

false alarm (cost 7-10).

3. Data Loading and Inspection

Due to the large training file size (1.22 GB), we sampled 50,000 rows for analysis. After merging all files:

Merged dataset: 50,000 rows x 117 columns

Unique vehicles in sample: 10,231

Observations per vehicle: mean 4.9 (range 1-29)

time_step range: 0 to 501.4 (mean 151.1, std 101.7)

4. Timestamp Processing

The time_step column represents operational time units since Component X installation - NOT calendar dates or clock time. Each vehicle has its own independent timeline.

Features Created

- lifecycle_progress: normalized position in vehicle observation (0.0 to 1.0) - useful for degradation tracking
- obs_number: sequential observation count per vehicle
- hour_bin, day_bin, is_weekend: artificial cyclic bins (time_step % 24 and % 168). NOTE: these do NOT represent real time-of-day. Created for assignment requirement.

5. Visualization and Distribution Analysis

5.1 Correlation Analysis

Top features correlated with failure (absolute Pearson correlation):

- length_of_study_time_step: 0.127 (shorter observation = more likely failed)
- Spec_0: 0.079 (certain truck types fail more)
- 459_15: 0.069 (histogram bin 15 of variable 459)
- Spec_3: 0.068, Spec_6: 0.066

All correlations are weak (<0.13), indicating non-linear models will be needed.

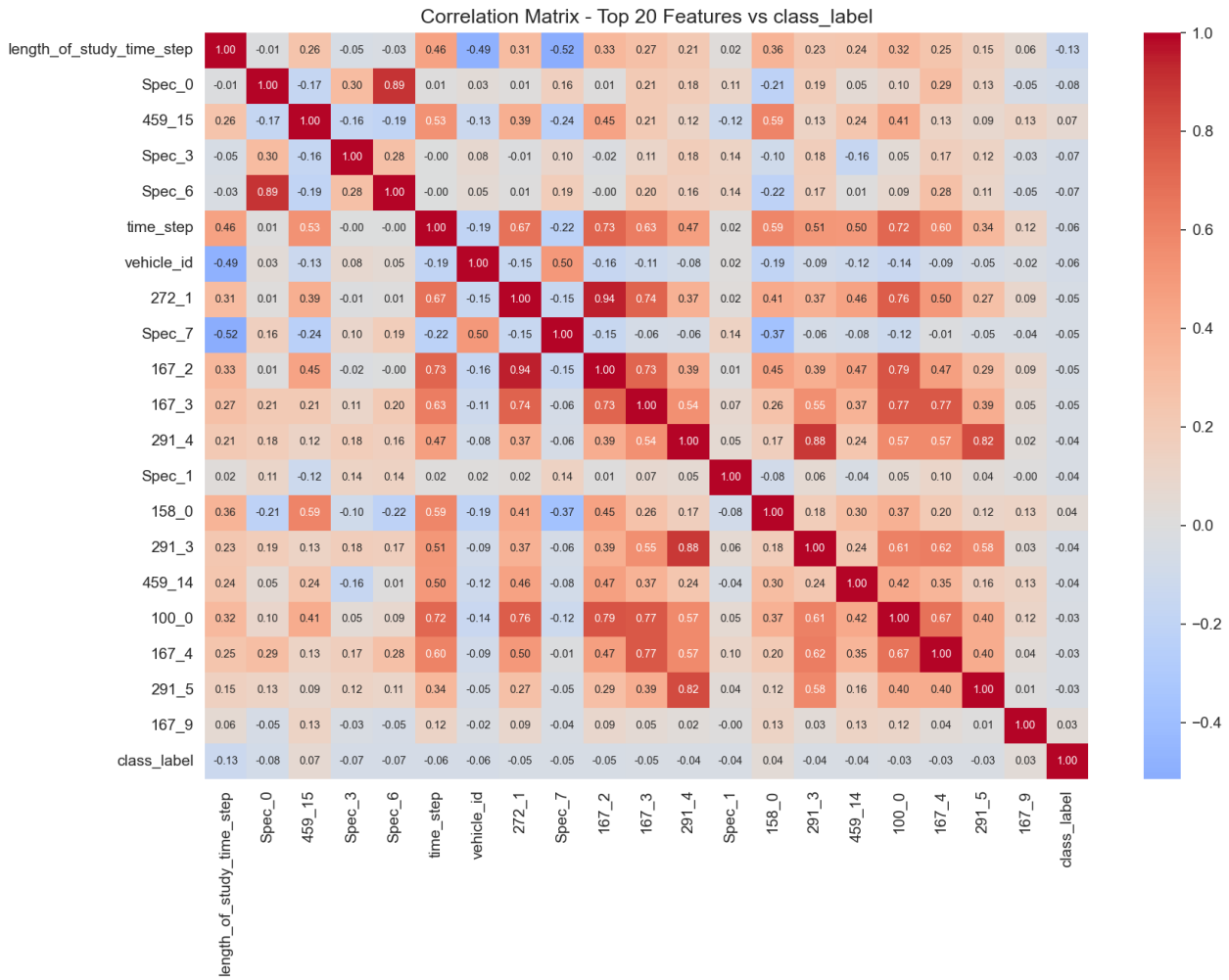


Figure 1: Correlation heatmap of top 20 features vs class_label. All individual correlations are weak (<0.13).

5.2 Feature Distributions

Histograms show highly skewed distributions typical for accumulative sensor data, with many near-zero values and long right tails.

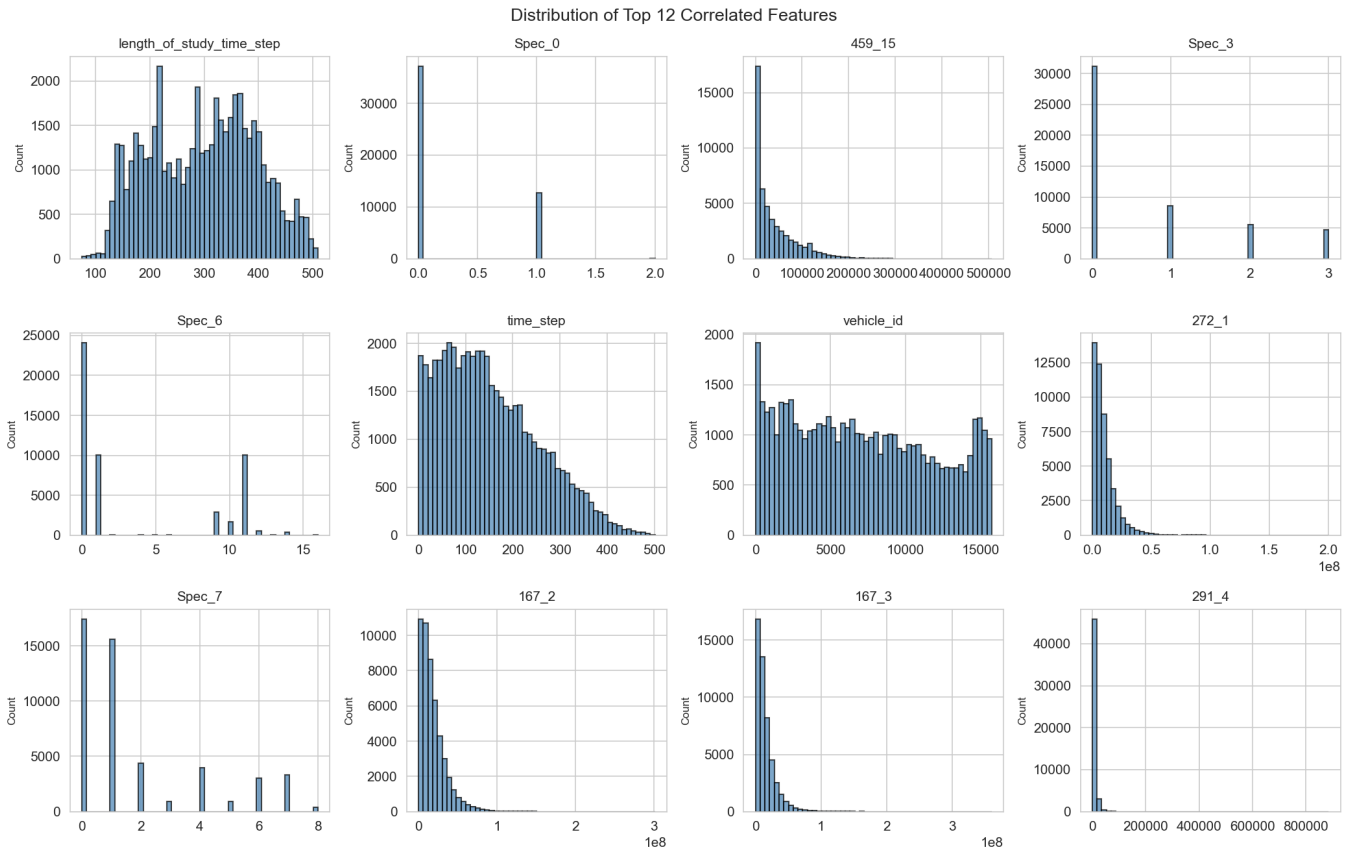


Figure 2: Distribution of top 12 correlated features. Most show right-skewed distributions.

5.3 Time Series Analysis

Numerical counters increase monotonically over time. Rolling means (12-observation window) reveal smooth trends useful for detecting degradation patterns.

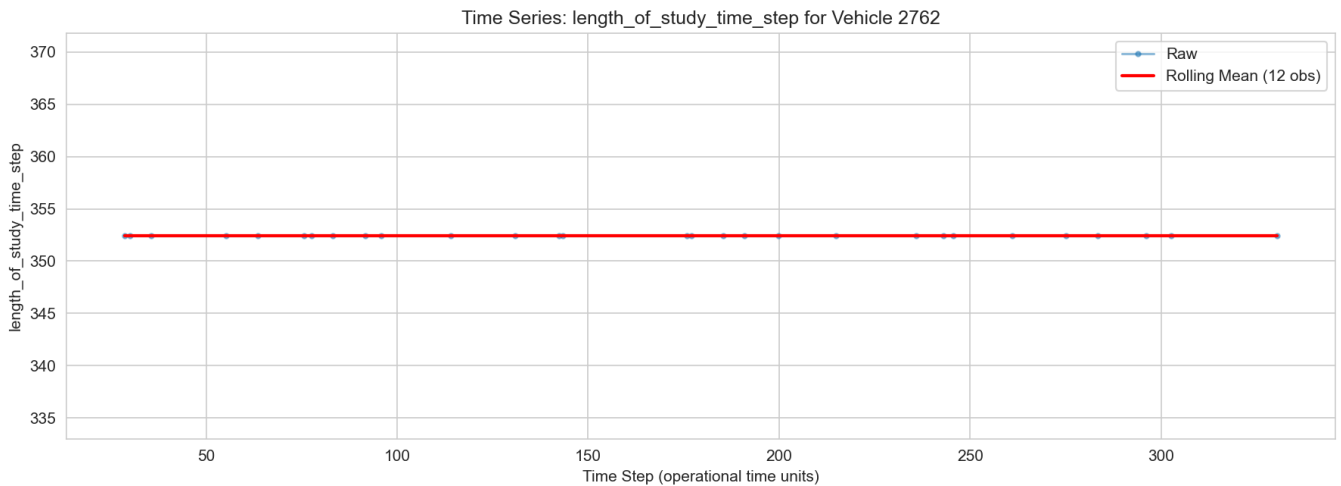


Figure 3: Time series of top feature for a sample vehicle with 12-observation rolling mean overlay.

5.4 Temporal Patterns

Average feature values across artificial hour/day bins show some variation, though these patterns are artifacts of the binning, not real temporal cycles.

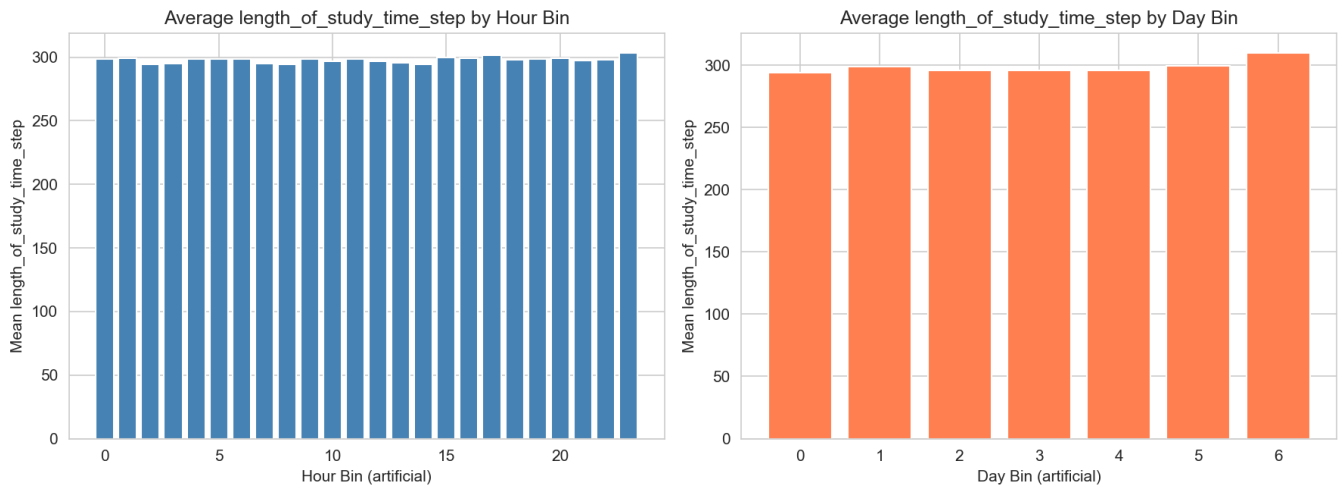


Figure 4: Average feature values by artificial hour and day bins (derived from time_step, not real clock time).

5.5 Class Distribution

Dataset imbalance: 9,118 no-repair vs 1,113 repaired vehicles (8.2:1 ratio in sample). Failed vehicles tend to have shorter study lengths.

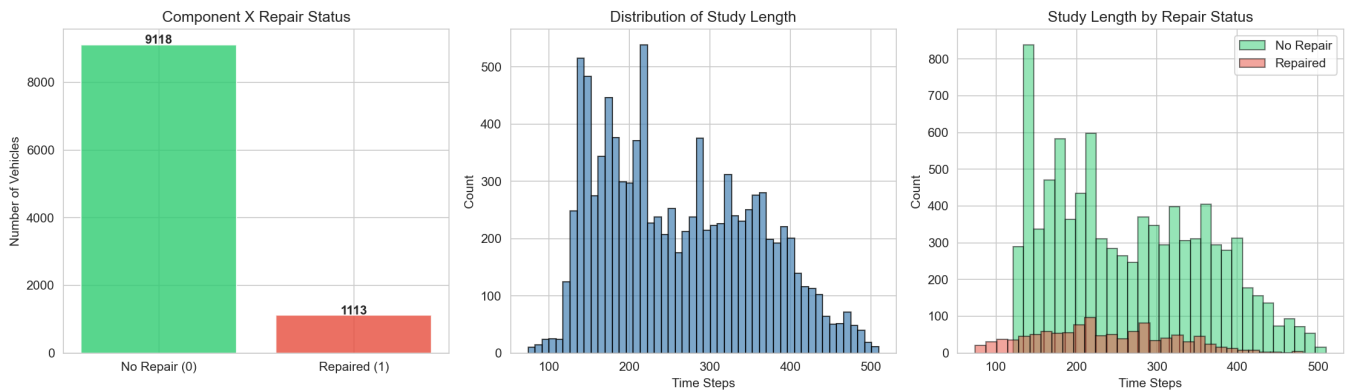


Figure 5: (Left) Repair status distribution. (Center) Study length distribution. (Right) Study length by repair status.

6. Handling Missing Values and Outliers

6.1 Missing Values

104 of 105 sensor columns had missing values, all below 1%. Causes (per documentation): sensor not installed, ECU not configured to log, or communication issues.

Highest: variable 291 (all bins) at 0.92%.

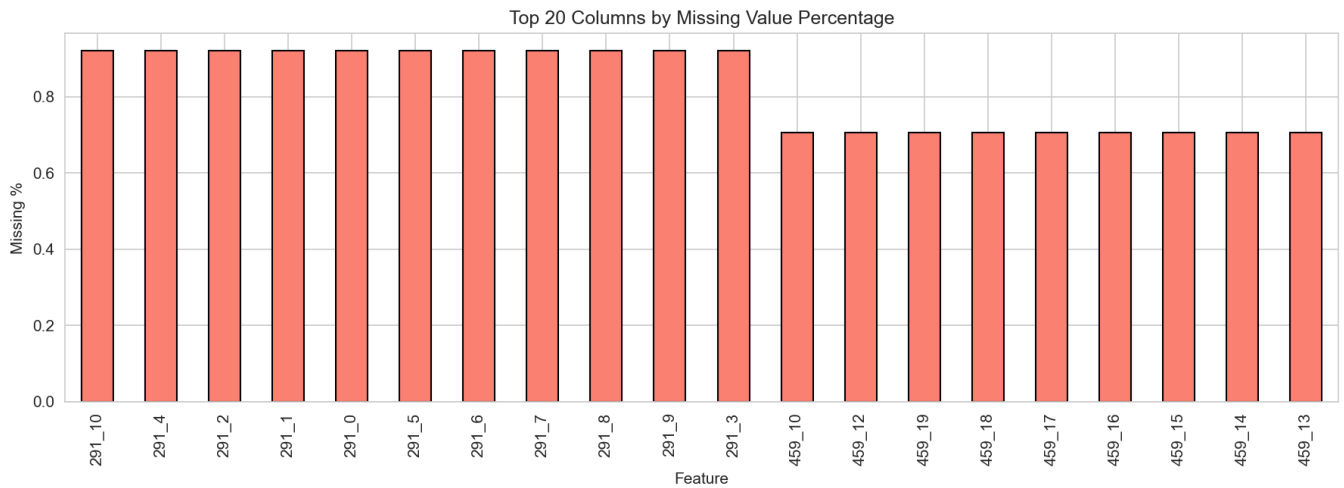


Figure 6: Missing value percentage by column. All features have <1% missing values.

Imputation Strategy

1. Forward-fill within each vehicle (carry last known reading)
2. Backward-fill for NaNs at the start
3. Column median for remaining NaNs

Result: 0 missing values after imputation.

6.2 Outlier Detection

IQR method detected outliers in all 105 sensor features. Top: 459_16 (9,563), 370_0 (9,540), 167_9 (8,853). Outliers were capped at IQR boundaries rather than removed, preserving potential failure signals.

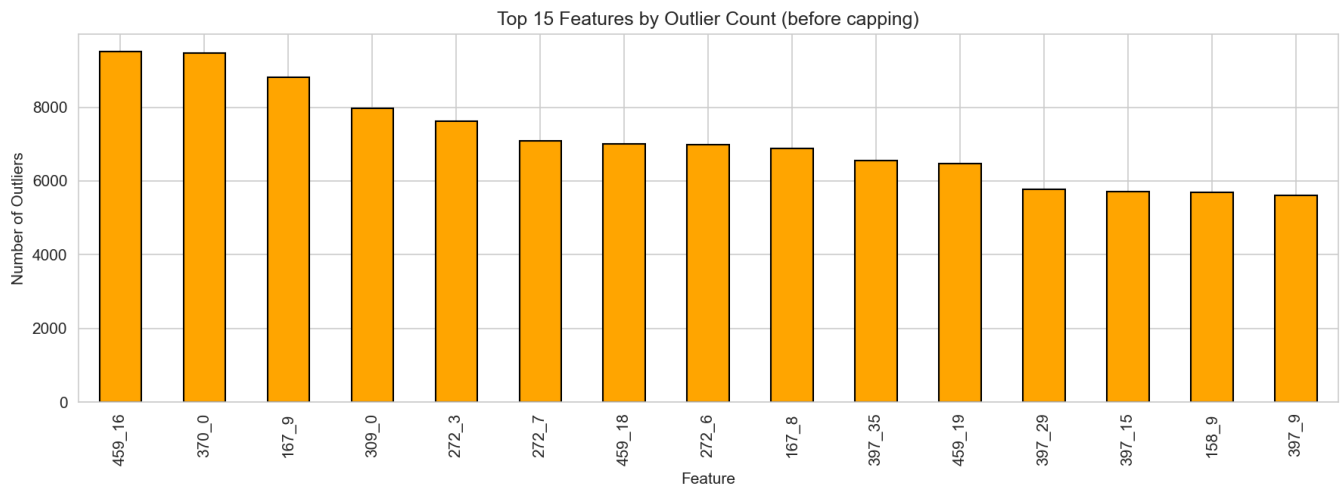


Figure 7: Top 15 features by outlier count before IQR capping.

7. Feature Engineering

7.1 Lag Features

lag_1, lag_2, lag_3 for top sensor (459_15): what the sensor read 1, 2, 3 observations ago. Helps models detect trends. First observations filled with current value.

7.2 Rolling Means

Windows of 3, 6, 12 observations for sensor 459_15. Captures short/medium/long-term trends.

7.3 Time-of-Day Category

Categories (morning/day/evening/night) from hour_bin. Artificial - not real clock time.

7.4 Rate of Change

Consecutive difference for sensor 459_15. Captures acceleration/deceleration of degradation.

Final Dataset

Processed dataset: 50,000 rows x 130 columns

- 105 original sensor features
- 2 identifiers (vehicle_id, time_step)
- 10 TTE/specification features
- 13 engineered features

Saved as processed_data.csv (60.8 MB).

8. Conclusions

1. The SCANIA Component X dataset is a complex real-world multivariate time series with 14 anonymized sensor variables (6 histograms + 8 counters) from 33,000+ trucks.
2. Correlations with failure are weak (<0.13) - neural networks and non-linear models will be essential for effective prediction.
3. Significant class imbalance (~9.6% failure rate) requires special handling (oversampling, class weights, or cost-sensitive learning).
4. Missing values are minimal ($<1\%$) and handled with time-aware imputation.
5. time_step represents operational time, not calendar time, enabling lifecycle-based analysis but limiting traditional temporal feature extraction.
6. The asymmetric cost matrix (false negative: 200-500, false positive: 7-10) must be central to model design and evaluation.

The processed dataset is ready for neural network development in Assignment 2.

9. References

1. Kharazian, Z., Lindgren, T., Magnusson, S., Steinert, O., & Andersson Reyna, O. (2025). SCANIA Component X dataset: a real-world multivariate time series dataset for predictive maintenance. Scientific Data, 12, 493.
2. IDA 2024 Industrial Challenge: Developing an Effective Predictive Model for Imminent Component X Failures in Heavy-Duty Scania Trucks.
3. Dataset: <https://researchdata.se/en/catalogue/dataset/2024-34>
4. GitHub Repository: <https://github.com/JaboDabo/SCANIA-Compenent-X-AI-NN->